SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I K·Shaik Nasrin *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K·Shaik Nas.

18/20

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightag e	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginnin g
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understandi ng; major misconcepti ons
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organizatio n, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understan d

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CS1703 - Artificial Intelligence TEST

1) Smart Home Automation System

A smart home Automation system is an intelligent system that automatically controls sensors, temperature and security by using the sensors and user preferences. It receives the information from the environment, processes it and takes actions.

Types of Agents:

- Simple Reflex Agent
- Model Based Aged
- Goal Based Agent
- Utility Based Agent

* Simple Reflex Agent

- These type of agent takes input from the environment and take actions

- These dont have any previous memory or actions stored in them

for example :

motion based light (ON / OFF)

Advantages :

- Fast and easy to implement
- suitable for simple tasks to perform the goals

Disadvantages :

- It cannot remember the previous events
- It cannot learn the user preferences

* Model Based Agent

- These type of agents keep an internal mode of environment and remember the previous states

If the user or owner leaves the home it switch off lights, lock the doors activate security automatically

Advantages

- uses memory for making decision
- used well in changing enviroments

Disadvantages

- very complex than a simple reflex agent.

* Goal Based Model

- These type of models takes an action to acheive a specific goal

Advantages

- Makes intelligent decisions
- helps in finishing off a specific goal

Disadvantages

- Requires planning, so it take time and power example :

If temperature is <25°c, switch off Air conditioner

* Utility Based Agent

These type of Agents work on the basis of utility factors ~~&~~ It chooses action that has the high and overall benefit of most factors utility factors include - user comfort, electricity usage etc..

example :

The Air conditioning should be such that electricity usage should be under some limit

Advantages :

- helps in producing overall performance
- optimizes electricity energy consumption

Disadvantages

- utility calculation is difficult and time consuming

Thus for smart house automation system, it should be a hybrid of Model Based, Goal Based, and Utility Based Agents to produce best results.

Introduction

A Robot must find the exit of an unknown maze without prior knowledge. Uniformed search algorithms like uniform cost search (UCS) and Iterative Deepening Search (IDS) help solve this Problem

* **UCS** (Uniform Cost Search)

working -

- expands the node with lowest path cost first
- uses a priority queue

Advantages -

• complete
• Finds least cost optimal path.
• suitable when path cost differ

Disadvantages -

• High memory usage

- can be slow for large ma

complexity

- Time - $O(b^{1+[C^*/\varepsilon]})$
- Space - same as order

* IDS (Iterative Deepening Search)

- Working
  - Repeatedly performs depth - limited search, Increases depth limit one level at a time

- Advantages
  - Low memory requirement
  - Complete
  - suitable when solution depth is unknown

- DisAdwantages
  - repeats exploration of upper levels
  - Not optimal when cost vary

complexity

- Time-$O(b^d)$
- space-$O(bd)$

where,

b = branching factor

d = depth of solution

* Relation to Intelligent Agent

- **Goal Based Agent**
  - searches for the maze exit
  - Plans to take actions to acheive goals
- **Model Based Agent**
  - Builds map while exploring
  - avoids revisiting explored path

* Effect of Agent design
  - Better environment models improve navigation

- Planning reduces unnecessary movement
- Memory helps avoid repeated exploration

* <u>Best Combination</u>

A Goal Based Agent with UCS (uniform cost search is most effective

<u>Justification</u>

- The goal based agent focuses on reaching the exit
- The UCS gaurentees the lowest cost path
- If all paths have equal cost and memory is limited, DFS is a good alternative

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